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neural network models in forecasting and decision making with application

نماذج الشبكات العصبية في التنبؤ و اتخاذ القرارات مع التطبيق

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Abstract:

This study is a theoretical attempt to explore the topic of intelligent models in the field of administrative sciences. In this article, we highlight the importance of artificial neural networks as an intelligent system in the domain of management and business administration. The focus is placed on the forecasting function within management by developing and applying a predictive model that simulates the operational reality of internet service agencies in the Algerian market under study.

Keywords: Intelligent models, Neural networks , Modeling, Forecasting.
JELClassificationCodes: C23, C01, B11

ملخص:

تُعدّ هذه الدراسة محاولةً نظريةً لاستكشاف موضوع النماذج الذكية في مجال العلوم الإدارية. في هذه المقالة، نُسلّط الضوء على أهمية الشبكات العصبية الاصطناعية كنظام ذكي في مجال الإدارة وإدارة الأعمال. يُركّز البحث على وظيفة التنبؤ في الإدارة من خلال تطوير وتطبيق نموذج تنبؤي يُحاكي الواقع التشغيلي لوكالات خدمات الإنترنت في السوق الجزائرية قيد الدراسة

كلمات مفتاحية: نماذج ذكية، شبكات عصبية،نمذجة، تنبؤ.
تصنيفات JEL : C23، C01، B11

1. INTRODUCTION:

In today's highly competitive and rapidly evolving landscape, organizations distinguished by intelligence and strategic insight are better positioned to deliver exceptional value to their stakeholders. These organizations not only produce high-quality goods and services but also

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demonstrate a sustained ability to create and lead in value delivery through continuous innovation and responsiveness. Strategic decision-making grounded in organizational intelligence has become a key driver of success and long-term sustainability.

To improve performance and decision-making, contemporary enterprises have increasingly leveraged digital technologies, particularly computing systems, the Internet, and the broader web ecosystem. This evolution has given rise to the field of Business Intelligence (*BI, a comprehensive framework that integrates analytical tools, data infrastructures, information systems, and performance management methodologies)*(Chen, Chiang, & Storey, 2012). BI encompasses a wide array of technologies including data warehouses, mining algorithms, real-time analytics, and visualization platforms, all of which aim to support informed and data-driven decisions (Shollo & Galliers, 2016).

As business environments become more volatile, data-rich, and technologically integrated, BI systems now rely heavily on high-performance computing and global information networks. This allows organizations to process large volumes of structured and unstructured data for strategic insights. The complexity of modern decision-making thus necessitates not only access to information but also advanced tools capable of interpreting and modeling it effectively.

This research investigates one such advanced tool: expert systems, with a particular focus on Artificial Neural Networks (ANNs). The study seeks to answer the following overarching research question: ***To what extent do expert systems, particularly artificial neural networks, enhance administrative decision-making in the domains of management and business administration?***

To address this question, we explore the following sub-questions:

1. What role do expert systems play in supporting managerial decision-making processes?
2. What is the current state of expert system adoption in management and business administration?
3. How can artificial neural networks be used to construct predictive models as a form of expert system support?

By examining these questions, the study aims to contribute to the literature on intelligent decision-support systems in organizational contexts, and to assess the effectiveness of ANNas a forecasting tool in a real-world business environment.

2. Theoretical Framework: The Growing Role of Artificial Intelligence in Modern Business Management

This section highlights the growing importance of advanced technologies, particularly artificial intelligence, in modern business management. In the current era of rapid technological advancement, the integration of artificial intelligence (AI) into business practices has become not only a competitive advantage but a strategic necessity. Organizations across industries are increasingly relying on intelligent systems to improve efficiency, accuracy, and responsiveness. Among the most impactful AI technologies in the domain of business management are artificial neural networks (ANNs) and expert systems, both of which contribute significantly to data-driven decision-making and process automation.

2.1 Artificial Neural Networks (ANNs): Artificial neural networks are computational models inspired by the structure and functioning of the human brain. They consist of layers of interconnected nodes, or "neurons", that process and transmit information. These models are particularly well-suited for identifying hidden patterns in large datasets, managing uncertainty, and modeling complex, non-linear relationships.

In business management, ANNs are widely applied in areas such as:

- Demand forecasting, where they help predict future customer needs based on historical data;
- Sales prediction, allowing companies to optimize inventory and distribution;
- Strategic planning, by analyzing trends and simulating scenarios.

Their ability to "learn" from data makes them ideal for environments characterized by variability and high information flow. Once trained, ANNs can process new inputs quickly, offering real-time insights and predictive analytics that support effective managerial decisions. (Hamidane & Hamidane, 2018)

2.2 Expert Systems : Expert systems are a class of AI technologies that simulate the reasoning and decision-making abilities of human experts. They operate based on a structured knowledge base containing domain-specific facts and rules, and an inference engine that applies logical reasoning to arrive at conclusions or recommendations.

These systems are especially valuable in contexts requiring consistency, objectivity, and speed in decision-making. In business environments, expert systems are used for: Problem diagnosis, such as identifying inefficiencies in production systems or faults in operations; Resource optimization, where they propose solutions for better allocation of labor, capital, or materials; Decision support, providing structured advice in fields such as finance, human resources, and logistics. Their rule-based logic ensures transparent justifications for decisions, which is beneficial in regulated industries or organizations with high compliance standards.

2.3 Integration of AI in Business Management: The combined use of ANNs and expert systems significantly enhances organizational performance. AI technologies allow firms to analyze large volumes of data, detect trends, and respond quickly to internal and external changes. This digital transformation fosters a more agile and proactive approach to management. (Mantri & Mishra, 2023)

Furthermore, the integration of AI contributes to: Improved strategic alignment, by offering managers deeper insight into operational and market dynamics; Enhanced customer satisfaction, through better service personalization and responsiveness; Cost reduction, via automation of repetitive tasks and optimization of business processes.

As AI continues to evolve, hybrid systems that merge the strengths of neural networks (flexibility and pattern recognition) with expert systems (explicit knowledge reasoning) are being explored to address even more complex management challenges. The future of business management thus lies in leveraging these technologies not only for operational efficiency but also for strategic innovation.

3. METHODOLOGY: In this section, we apply the **Multi-Layer Perceptron (MLP)** neural network model due to its widespread use among researchers for modeling and forecasting time series data, as well as its strong predictive performance. We work with eight time series, and for each one, we aim to construct a suitable neural network model for forecasting.

Input specification : Given the nature of the available data, observations of Internet subscription levels in each agency, the input to each neural network model is a single time series representing subscription levels in the corresponding agency.

The study sample consists of the agencies of companies investing in Internet service provision in Algeria :

- **Algerie Télecom** (Saida and Maghnia agencies)
- **Mobilis** (Saida and Maghnia agencies)
- **Ooredoo** (Saida and Maghnia agencies)
- **Djezzy** (Saida and Maghnia agencies).

4. RESULTS:

Analysis phase:Neural Network Construction for the Time Series Under Study: In this phase, we analyze and build the neural networks by dividing the observations into three working groups, each playing a distinct role in the network's training and evaluation.

1 Analysis of Mobilis – Maghnia Agency: By compiling the time series data representing internet subscriptions at the Mobilis – Maghnia Agency (a total of 48 observations), the dataset was randomly divided into three primary groups to support the process of network modeling and forecasting. This is illustrated in the following table.

Table No. 01: results of the Analysis of the Internet subscription network of Mobilis – Maghnia agency

Groups	AIMM	
	N° of observations	**Percentage**
Training group:	30	62.50%
Validation group:	10	20.83%
Testing group:	08	16.67%
Total Observations	48	100%

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

2. **Analysis of Mobilis – Saida Agency**

48 observations were split as follows:

Table No. 02: results of the Analysis of the Internet subscription network of Mobilis – Saida agency

Groups	**AIMS**	
	N° of observations	**Percentage**
Training group:	29	60.41%
Validation group:	11	22.91%
Testing group:	08	16.68%
Total Observations	48	100%

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

3. **Analysis of Djezzy – Maghnia Agency**

48 observations were split into:

Table No. 03: results of the Analysis of the Internet subscription network of Djezzy –Maghnia agency

Groups	AIDM	
	N° of observations	**Percentage**
Training group:	34	70.84%
Validation group:	7	14.58%
Testing group:	7	14.58%
Total Observations	48	100%

Source: Compiled by the researchers using the Alyda NeuroIntelligennce program.

4. Analysis of Djezzy – Saida Agency

48 observations were split into:

Table No. 04: results of the Analysis of the Internet subscription network of Djezzy – Saida agency

Groups	AIDS	
	N° of observations	**Percentage**
Training group:	29	60.42%
Validation group:	10	20.83%
Testing group:	09	18.75%
Total Observations	48	100%

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

5.Analysis of Ooredoo – Maghnia Agency

36 observations were split into:

Table No. 05: results of the Analysis of the Internet subscription network of Ooredoo- Maghnia agency

Groups	AIOM	
	N° of observations	**Percentage**
Training group:	24	66.66%
Validation group:	7	19.44%
Testing group:	5	13.89%
Total Observations	36	100%

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

6.Analysis of Ooredoo – Saida Agency

36 observations were split into:

Table No. 06: results of the Analysis of the Internet subscription network of Ooredoo – Saida agency

Groups	AIOS	
	N° of observations	**Percentage**
Training group:	20	55.55%
Validation group:	09	25.00%
Testing group:	07	19.45%
Total Observations	36	100%

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

7. Analysis of Algérie Télécom – Maghnia Agency

60 observations were split into:

Table No. 07: results of the Analysis of the Internet subscription network of Algérie Télécom – Maghnia agency

Groups	AIATM	
	N° of observations	**Percentage**
Training group:	38	63.33%
Validation group:	12	20.00%
Testing group:	10	16.67%
Total Observations	60	100%

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

8. Analysis of Algérie Télécom – Saida Agency

60 observations were split into:

Table No. 08: results of the Analysis of the Internet subscription network of Algérie Télécom – Saida agency

Groups	AIATS	
	N° of observations	**Percentage**
Training group:	35	58.33%
Validation group:	15	25.00%
Testing group:	10	16.67%
Total Observations	60	100%

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

II. Neural Network Model Estimation: At this stage of the research, several candidate models will be estimated to represent the time series under investigation. These models will then be evaluated and compared using a set of predefined criteria.

2-1- Estimating the Neural Network Architecture for Mobilis – Maghnia Agency: As presented in the table, two neural network architectures were proposed for forecasting internet subscriptions at the Mobilis – Maghnia Agency.

Table 9: Results of Neural Network Architecture Estimation – Mobilis Maghnia

Estimated Model	Training Error	Test Error	AIC Criterion	R-Squared
MLP 1_3_1	0.004431	0.000971	-603.00	0.999
MLP 1_7_1	0.001111	0.000966	-489.85	0.988

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

Based on criteria such as the AIC and error minimization, the optimal architecture is **MLP 1_7_1**, which consists of three layers:

- Input layer: 1 processing unit
- Hidden layer: 7 processing units
- Output layer: 1 processing unit.

2.2. Estimating the Architecture for Mobilis – Saida Agency: Three models were proposed.

Table 10: Results of Neural Network Architecture Estimation – Mobilis Saida

Estimated Model	Training Error	Test Error	AIC Criterion	R-Squared
MLP 1_1_1	0.002222	0.0003519	-606.90	0.990
MLP 1_4_1	0.001299	0.0000121	-1012.02	0.970
MLP 1_7_1	0.001312	0.0007000	-1000.01	0.899

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

The optimal network is **MLP 1_4_1**:

- Input layer: 1 unit
- Hidden layer: 4 units
- Output layer: 1 unit

2.3.Estimating the Architecture for Djezzy – Maghnia Agency

Five models were proposed.

Table 11: Neural Network Architecture Estimation – Djezzy Maghnia

Estimated Model	Training Error	Test Error	AIC Criterion	R-Squared
MLP 1_2_1	0.00121212	0.000521	-120.09	0.999
MLP 1_5_1	0.00012111	0.000111	-444.92	0.999
MLP 1_7_1	0.00823445	0.000123	-310.00	0.987
MLP 1_8_1	0.00213450	0.000229	-253.09	0.898
MLP 1_9_1	0.00600120	0.000211	-309.99	0.990

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

Best model: **MLP 1_5_1**

2.4. **Estimating the Architecture for Djezzy – Saida Agency**

Three models were considered.

Table 12: Neural Network Architecture Estimation – Djezzy Saida

Estimated Model	Training Error	Test Error	AIC Criterion	R-Squared
MLP 1_1_1	0.00180	0.000010	-1201.03	0.999
MLP 1_7_1	0.00213	0.000312	-1129.00	0.999
MLP 1_9_1	0.00711	0.000231	-999.37	0.999

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

Best model: **MLP 1_1_1**

2.5.Estimating the Architecture for Ooredoo – Maghnia Agency

Two models were proposed.

Table 13: Neural Network Architecture Estimation – Ooredoo Maghnia

Estimated Model	Training Error	Test Error	AIC Criterion	R-Squared
MLP 1_2_1	0.001298	0.000016	-1213.54	0.999
MLP 1_9_1	0.000120	0.000001	-1108.00	0.999

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

Best model: **MLP 1_9_1.**

2.6. Estimating the Architecture for Ooredoo – Saida Agency

Two models were proposed.

Table 14: Neural Network Architecture Estimation – Ooredoo Saida

Estimated Model	Training Error	Test Error	AIC Criterion	R-Squared
MLP 1_1_1	0.000421	0.0001212	-901.76	0.999
MLP 1_3_1	0.000100	0.0001123	-1321.98	0.999

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

Best model: **MLP 1_3_1**

2.7.Estimating the Architecture for Algérie Télécom – Maghnia Agency

Three models were proposed.

Table 15: Neural Network Architecture Estimation – Algérie Télécom Maghnia

Estimated Model	Training Error	Test Error	AIC Criterion	R-Squared
MLP 1_3_1	0.001221	0.000454	-113.16	0.998
MLP 1_5_1	0.001123	0.000444	-134.76	0.999
MLP 1_7_1	0.001111	0.000512	-138.09	0.997

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

Best model: **MLP 1_5_1**

2.8. Estimating the Architecture for Algérie Télécom – Saida Agency

Four models were proposed.

Table 16: Neural Network Architecture Estimation – Algérie Télécom Saida

Estimated Model	Training Error	Test Error	AIC Criterion	R-Squared
MLP 1_2_1	0.003219	0.0007880	-257.54	0.949
MLP 1_6_1	0.002213	0.0001212	-341.90	0.999
MLP 1_7_1	0.007777	0.0005410	-333.98	0.912
MLP 1_8_1	0.003112	0.0005666	-120.02	0.987

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

Best model: **MLP 1_6_1**

III. Forecasting Using the Optimal Neural Network Models :

At this stage of the research, internet subscriptions for the agencies of the operators in the Algerian market under study will be forecasted using the neural networks previously built for each agency. The forecasting covers the period from January 2017 to December 2017.

The following table presents the forecasting results:

Table No. 17: Forecasting Results Using the Optimal MLP Neural Network Models Constructed

Month	AIATM	AIATS	AIMM	AIMS	AIDM	AIDS	AIOM	AIOS
January	500.56	701.56	389.89	421.56	302.87	401.18	389.73	499.22
February	531.56	712.86	433.33	489.71	400.56	444.90	399.49	517.52
March	537.09	753.98	459.10	499.99	452.98	465.12	434.91	555.79
April	540.67	782.09	500.66	511.45	498.09	500.92	480.01	570.12
May	542.98	799.12	545.07	560.90	521.48	571.77	513.16	603.33
June	556.90	800.66	580.80	577.71	589.67	590.00	532.59	621.75
July	570.88	823.16	596.99	593.33	617.88	611.98	627.91	671.00
August	601.16	831.11	609.12	630.99	695.91	643.34	677.06	698.11
September	622.32	846.14	623.90	642.67	711.39	667.67	752.61	707.64
October	630.00	857.90	643.98	672.22	741.19	712.09	782.72	745.54
November	647.69	878.00	670.11	701.01	769.01	768.98	802.50	788.99
December	681.11	881.67	689.89	732.99	799.81	792.17	839.14	827.32

Source: Compiled by the researchers using the Alyda NeuroIntelligence program.

Based on the forecasting results shown in the table, and according to the data from the centers used for testing, it appears that these results are relatively close to the actual demand for their service products, namely internet subscriptions. This suggests that there is a scientific basis for employing such models in management and administrative decision-making.

After selecting the best-performing neural network models for each time series based on performance metrics (training error, test error, AIC, R^2), the final phase involves using these models to generate **forecasted values**. These forecasts represent future Internet subscription levels in the selected agencies based on historical data patterns.

- **Forecasting for Mobilis – Maghnia Agency :**The **MLP 1_7_1** model was used to predict future values. The forecasted values closely matched the actual test values, indicating a high level of accuracy.
- **Forecasting for Mobilis – Saida Agency :**The **MLP 1_4_1** model generated the forecasts. The predicted subscription levels demonstrate excellent fit with minimal deviation.
- **Forecasting for Djezzy – Maghnia Agency :**Using the **MLP 1_5_1** model, the network produced reliable forecasts, showing strong alignment with actual observed values.
- **Forecasting for Djezzy – Saida Agency :**The selected **MLP 1_1_1** model yielded highly accurate predictions, reflecting the strength of the model’s simplicity in capturing the underlying pattern.
- **Forecasting for Ooredoo – Maghnia Agency :**The **MLP 1_9_1** model successfully anticipated future subscription behavior, validating the model's structure and training.

- **Forecasting for Ooredoo – Saida Agency :**The **MLP 1_3_1** architecture demonstrated excellent predictive ability, maintaining consistency across validation and testing data.
- **Forecasting for Algérie Télécom – Maghnia Agency :**The **MLP 1_5_1** network accurately forecasted the Internet subscription levels, confirming its selection as the optimal model.
- **Forecasting for Algérie Télécom – Saida Agency :**Forecasting with the **MLP 1_6_1** model generated precise and dependable results, affirming the effectiveness of neural networks in time series prediction.

Each of the optimal models showed strong performance in both training and testing phases, highlighting the capacity of **Artificial Neural Networks (ANNs)** to effectively model and forecast time series data in the context of Internet service provision.

5. CONCLUSION:

In light of the growing complexity and volatility of today's business environment, organizations are increasingly seeking intelligent tools to support strategic and operational decision-making. This study aimed to explore the applications of Artificial Neural Networks **(ANNs)** as expert systems within the fields of management and business administration, particularly in the area of forecasting.

Through an empirical case study involving Algerian mobile Internet service providers (Mobilis, Djezzy, Ooredoo, and Algérie Télécom), we modeled and predicted time series data related to Internet subscription levels using Multi-Layer Perceptron (MLP) networks.

The main conclusions are as follows:

1. Artificial neural networks have proven to be effective expert systems capable of learning from historical data and generating reliable forecasts for decision-making purposes.
2. The MLP models demonstrated high predictive accuracy across all tested agencies, with minimal errors and high R^2 values, confirming their suitability for modeling business data with temporal structures.
3. The integration of intelligent systems, such as ANNs, in managerial forecasting can significantly enhance the quality of decisions, especially in sectors characterized by demand variability and service competition.

Based on the findings of this study, it is recommended that decision-makers in internet service provider companies adopt artificial intelligence tools—particularly neural networks—as effective instruments for forecasting demand and optimizing resource allocation. To ensure successful implementation, companies should invest in training both managers and technical staff in the use and integration of AI within business processes. Additionally, expanding the application of expert systems to other key managerial domains, such as customer relationship management, pricing strategies, and service quality prediction, could further enhance organizational performance. Future research may also benefit from exploring hybrid models that combine artificial neural networks with other intelligent techniques, such as genetic algorithms, to strengthen decision-making capabilities in increasingly complex environments.

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